Appendix: Theoretical Analysis of the Augmented Synthetic Historical Control Estimator (ASHC)

1 Setup and Assumptions

For the purposes of self-containment, let $\mathcal{T} = \{1, 2, \ldots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \ldots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$. The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period. SHC by Chen et al. [2024] constructs $N = T_0 - m - n + 1$ overlapping temporal segments of length m. Each segment $\widetilde{\mathbf{y}}_{[i,i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series, forming the donor matrix

$$\widetilde{\mathbf{Y}}_{\text{pre}} := \begin{bmatrix} \widetilde{\mathbf{y}}_{[1,m]} & \widetilde{\mathbf{y}}_{[2,m+1]} & \cdots & \widetilde{\mathbf{y}}_{[N,N+m-1]} \end{bmatrix} \in \mathbb{R}^{m \times N}.$$

Example 1.1. *If $T_0 = 60$ and we want the segment length to be $m = 12$ months to predict a post-treatment horizon of $n = 6$ months, $N = 60 - 12 - 6 + 1 = 43$, where each segment is 12 months long.*

Respectively, SHC and ASHC solve:

$$\mathbf{w}^{\text{SHC}} = \underset{\mathbf{w} \in \mathbb{R}^N}{\operatorname{argmin}} \quad \left\| \widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \quad \text{subject to } \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1.$$

$$\mathbf{w}^{\text{ASHC}} = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2\lambda} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} \mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \right\} \quad \text{s.t. } \sum_{i=1}^N w_i = 1.$$

Here, $\lambda > 0$ is the regularization parameter, $\varsigma > 0$ is a penalty parameter, and $\mathbf{C}_0 \in \mathbb{R}^{N \times k}$ contains the eigenvectors of $\mathbf{G} = \widetilde{\mathbf{Y}}^\top \widetilde{\mathbf{Y}}$ corresponding to eigenvalues less than a threshold $\tau > 0$.

1.1 Assumptions

Assumption 1.1 (Linearity with noise). *There exists $\mathbf{w}^* \in \mathbb{R}^N$ such that $\widetilde{\mathbf{y}} = \widetilde{\mathbf{Y}} \mathbf{w}^* + \boldsymbol{\eta}$, where $\boldsymbol{\eta} \in \mathbb{R}^m$ is a mean-zero noise vector.*

Assumption 1.2 (Bounded noise). *$\|\boldsymbol{\eta}\| \leq \delta$ for some $\delta > 0$.*

Assumption 1.3 (Smoothness of latent trend, from Chen et al. [2024]). *Let $\ell(t)$ denote the latent smooth component of the treated potential outcome. Suppose $\ell(t)$ is $H + 1$ times differentiable with $|\ell^{(H+1)}(t)| < \epsilon$ for all t and some $H \geq 3$.*

Assumption 1.4 (Operator norm bound). *$\|\widetilde{\mathbf{Y}}\| \leq \gamma$.*

Assumption 1.5 (Simplex feasibility). *$\mathbf{w}^{SHC} \in \Delta^{N-1} := \{\mathbf{w} \in \mathbb{R}^N : \sum w_i = 1, \ w_i \geq 0\}$.*

Lemma 1.1 (Deviation from SHC)**.** *Let* $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$. *Then*

$$\|\Delta\| \;\leq\; \frac{\gamma}{2\lambda + 2\lambda\varsigma\|\mathbf{C}_0\|^2}\,\epsilon_{SHC}.$$

where $\epsilon_{SHC} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ *and* $\|\mathbf{C}_0\|^2$ *is the operator norm of* $\mathbf{C}_0^\top \mathbf{C}_0$.

Proof. To analyze the ASHC optimization problem, recall the objective function:

$$J(\mathbf{w}) = \frac{1}{2\lambda}\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\text{SHC}}\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2, \tag{1.1}$$

subject to the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$.

To enforce this constraint, we form th Lagrangian:

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1), \tag{1.2}$$

where μ is a Lagrange multiplier ensuring feasibility of $\mathbf{w}$. We compute the gradient of $\mathcal{L}$ with respect to $\mathbf{w}$ by differentiating each term separately.

For the first term, define the residual vector:

$$r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}. \tag{1.3}$$

The fit term is then:

$$\frac{1}{2\lambda}\|r(\mathbf{w})\|^2 = \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w}). \tag{1.4}$$

We can see that this is a standard quadratic form. We begin by applying the Chain Rule for vector calculus. The chain rule for a composite function $f(g(\mathbf{w}))$ is given by:

$$\nabla_{\mathbf{w}} f(g(\mathbf{w})) = \left(\nabla_{g(\mathbf{w})} f\right)^\top \nabla_{\mathbf{w}} g(\mathbf{w})$$

In our case, the inner function is the residual vector $g(\mathbf{w}) = r(\mathbf{w})$, and the outer function is the squared norm $f(r) = \frac{1}{2\lambda} r^\top r$. First, let's find the gradient of the outer function with respect to the residual vector r.

$$\nabla_r \left(\frac{1}{2\lambda} r^\top r\right) = \frac{1}{2\lambda}\nabla_r \left(\sum_i r_i^2\right) = \frac{1}{2\lambda}\begin{pmatrix} 2r_1 \\ 2r_2 \\ \vdots \end{pmatrix} = \frac{1}{\lambda} r$$

This is a well-known result: the gradient of $\|r\|^2$ is $2r$. The $\frac{1}{2\lambda}$ is a constant scaling factor.

Next, we need to find the gradient of the inner function, $r(\mathbf{w})$, with respect to the weight vector $\mathbf{w}$. This is the Jacobian matrix of the transformation.

$$\nabla_{\mathbf{w}} r(\mathbf{w}) = \nabla_{\mathbf{w}}(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w})$$

Since $\widetilde{\mathbf{y}}$ is a constant vector with respect to $\mathbf{w}$, its gradient is zero. We only need to compute the gradient of $-\widetilde{\mathbf{Y}}\mathbf{w}$.

$$\nabla_{\mathbf{w}}(-\widetilde{\mathbf{Y}}\mathbf{w}) = -\widetilde{\mathbf{Y}}$$

This result comes from the general rule that for a linear function $A\mathbf{x}$, the gradient with respect to $\mathbf{x}$ is just the matrix A. Here, our linear function is a negative, so the gradient is $-\widetilde{\mathbf{Y}}$.

Now we plug the results from the above back into the chain rule formula.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \left(\nabla_{r(\mathbf{w})} \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w})\right)^\top \nabla_{\mathbf{w}} r(\mathbf{w})$$

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \left(\frac{1}{\lambda} r(\mathbf{w})\right)^{\top} (-\widetilde{\mathbf{Y}})$$

Using the property that $(A\mathbf{x})^{\top} = \mathbf{x}^{\top} A^{\top}$, we get:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{1}{\lambda} r(\mathbf{w})^{\top} (-\widetilde{\mathbf{Y}}) = -\frac{1}{\lambda} r(\mathbf{w})^{\top} \widetilde{\mathbf{Y}}$$

This is equivalent to:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{\lambda} \widetilde{\mathbf{Y}}^{\top} r(\mathbf{w})$$

Finally, we substitute the definition of the residual vector, $r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}$, back into the gradient expression.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{\lambda} \widetilde{\mathbf{Y}}^{\top} (\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w})$$

Distributing the negative sign gives us the final form:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{1}{\lambda} \widetilde{\mathbf{Y}}^{\top} (\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}})$$

Next, consider the deviation penalty term:

$$\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = (\mathbf{w} - \mathbf{w}^{\mathrm{SHC}})^{\top} (\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}). \tag{1.5}$$

Its gradient with respect to $\mathbf{w}$ is:

$$\nabla_{\mathbf{w}} \|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}). \tag{1.6}$$

Notice that the Hessian of this term is $2\mathbf{I}$, because differentiating again with respect to $\mathbf{w}$ gives:

$$\nabla^2_{\mathbf{w}} \|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2\mathbf{I}. \tag{1.7}$$

Here $\mathbf{I}$ is the identity matrix, representing curvature equally in all coordinate directions. Thus the $2\mathbf{I}$ in M directly reflects the Hessian of this quadratic deviation penalty.

For the Gram penalty term, rewrite it as:

$$\varsigma \|\mathbf{C}_0^{\top} \mathbf{w}\|^2 = \varsigma \mathbf{w}^{\top} \mathbf{C}_0 \mathbf{C}_0^{\top} \mathbf{w}. \tag{1.8}$$

Since $\mathbf{C}_0 \mathbf{C}_0^{\top}$ is symmetric, its gradient is:

$$\nabla_{\mathbf{w}} \varsigma \mathbf{w}^{\top} \mathbf{C}_0 \mathbf{C}_0^{\top} \mathbf{w} = 2\varsigma \mathbf{C}_0 \mathbf{C}_0^{\top} \mathbf{w}. \tag{1.9}$$

Finally, the affine constraint term differentiates as:

$$\nabla_{\mathbf{w}} \mu(\mathbf{1}^{\top} \mathbf{w} - 1) = \mu \mathbf{1}. \tag{1.10}$$

We now have the first-order condition from the gradient of the Lagrangian:

$$\frac{1}{\lambda} \widetilde{\mathbf{Y}}^{\top} (\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}}) + 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}) + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^{\top} \mathbf{w} + \mu \mathbf{1} = \mathbf{0}. \tag{1.11}$$

Expanding the deviation penalty term gives

$$2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}) = 2\mathbf{w} - 2\mathbf{w}^{\mathrm{SHC}}. \tag{1.12}$$

Substituting back, the condition becomes

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w} - \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w} - 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w} + \mu\mathbf{1} = \mathbf{0}. \tag{1.13}$$

Now collect all the terms involving $\mathbf{w}$ on the left-hand side:

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w} \tag{1.14}$$

and move all the other terms to the right-hand side:

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.15}$$

Thus we obtain the compact matrix form:

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.16}$$

Here the $2\mathbf{I}$ arises directly from the quadratic penalty $\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2$, whose gradient contributes $2\mathbf{w}$, corresponding in matrix form to $2\mathbf{I}\cdot\mathbf{w}$.

Next, define the matrix

$$M = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top. \tag{1.17}$$

M corresponds to the Hessian of the objective function $J(\mathbf{w})$ plus the identity-weighted contribution of the deviation penalty.

The solution $\mathbf{w}^{\mathrm{ASHC}}$ satisfies the first-order condition:

$$M\mathbf{w}^{\mathrm{ASHC}} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.18}$$

Evaluating the same matrix M at $\mathbf{w}^{\mathrm{SHC}}$, which is not necessarily the ASHC solution, gives:

$$M\mathbf{w}^{\mathrm{SHC}} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\mathrm{SHC}}. \tag{1.19}$$

Subtracting this equation from the previous one, we get

$$M\mathbf{w}^{\mathrm{ASHC}} - M\mathbf{w}^{\mathrm{SHC}} = \left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}\right) - \left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\mathrm{SHC}}\right). \tag{1.20}$$

Simplifying the right-hand side by canceling the $2\mathbf{w}^{\mathrm{SHC}}$ terms, we obtain

$$M(\mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}) = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\left(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\right) - \mu\mathbf{1}. \tag{1.21}$$

Define the deviation

$$\Delta = \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}, \tag{1.22}$$

and the residual vector

$$r = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}. \tag{1.23}$$

Since both $\mathbf{w}^{\mathrm{ASHC}}$ and $\mathbf{w}^{\mathrm{SHC}}$ satisfy the affine constraint $\mathbf{1}^\top\mathbf{w} = 1$, their difference Δ lies in the hyperplane orthogonal to $\mathbf{1}$:

$$\mathbf{1}^\top\Delta = 0. \tag{1.24}$$

Taking the inner product of both sides of the previous equation with Δ gives

$$\Delta^\top M\Delta = \frac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top r - \mu\underbrace{\Delta^\top\mathbf{1}}_{=0} = \frac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top r. \tag{1.25}$$

Here, the right-hand side is an inner product between the deviation vector Δ and the vector $\widetilde{\mathbf{Y}}^\top r$. Because inner products can be difficult to control directly, we apply the Cauchy–Schwarz inequality to bound it:

$$|\Delta^\top\widetilde{\mathbf{Y}}^\top r| \le \|\Delta\|\,\|\widetilde{\mathbf{Y}}^\top r\|. \tag{1.26}$$

Thus, $\Delta^\top M\Delta \le \frac{1}{\lambda}\|\Delta\|\,\|\widetilde{\mathbf{Y}}^\top r\|$. By assumption, the operator norm of $\widetilde{\mathbf{Y}}$ is bounded by γ, and the residual norm is $\epsilon_{\text{SHC}} = \|r\|$, so

$$\|\widetilde{\mathbf{Y}}^\top r\| \le \|\widetilde{\mathbf{Y}}\|\|r\| \le \gamma\epsilon_{\text{SHC}}. \tag{1.27}$$

Thus, $\Delta^\top M\Delta \le \frac{\gamma}{\lambda}\|\Delta\|\epsilon_{\text{SHC}}$. On the other hand, since M is positive definite, its smallest eigenvalue $\lambda_{\min}(M)$ guarantees $\Delta^\top M\Delta \ge \lambda_{\min}(M)\|\Delta\|^2$. Putting these two inequalities together, we sandwich the quadratic form: $\lambda_{\min}(M)\|\Delta\|^2 \le \Delta^\top M\Delta \le \frac{\gamma}{\lambda}\|\Delta\|\,\epsilon_{\text{SHC}}$.

The left-hand side is the lower bound derived from the smallest eigenvalue of the positive definite matrix M. The right-hand side is the upper bound derived from the Cauchy-Schwarz and operator norm inequalities. We can now discard the middle term to focus on the key inequality between the two bounds: $\lambda_{\min}(M)\|\Delta\|^2 \le \frac{\gamma}{\lambda}\|\Delta\|\,\epsilon_{\text{SHC}}$. Assuming $\|\Delta\| > 0$ (the trivial case where the weights are identical is already clear), we can divide both sides of the inequality by $\|\Delta\|$ to isolate the term of interest. $\lambda_{\min}(M)\|\Delta\| \le \frac{\gamma}{\lambda}\epsilon_{\text{SHC}}$. Rearranging this inequality to solve for $\|\Delta\|$ gives us an intermediate bound: $\|\Delta\| \le \frac{\gamma}{\lambda\lambda_{\min}(M)}\epsilon_{\text{SHC}}$. Next, we establish a lower bound for the smallest eigenvalue of the matrix M. Recall that the matrix M is defined as: $M = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top$. The smallest eigenvalue of a sum of positive semi-definite matrices is at least the sum of their smallest eigenvalues. Since the first term's smallest eigenvalue is non-negative, and the second term's is 2, we have:

$$\lambda_{\min}(M) \ge \lambda_{\min}\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\right) + \lambda_{\min}(2\mathbf{I}) + \lambda_{\min}(2\varsigma\mathbf{C}_0\mathbf{C}_0^\top)$$

$$\lambda_{\min}(M) \ge 0 + 2 + 2\varsigma\lambda_{\min}(\mathbf{C}_0\mathbf{C}_0^\top)$$

By the assumption that $\lambda_{\min}(\mathbf{C}_0\mathbf{C}_0^\top) \ge \|\mathbf{C}_0\|^2$, we get the lower bound for the smallest eigenvalue of M:

$$\lambda_{\min}(M) \ge 2 + 2\varsigma\|\mathbf{C}_0\|^2$$

Finally, we substitute this lower bound for $\lambda_{\min}(M)$ back into the intermediate bound for $\|\Delta\|$. A smaller value in the denominator leads to a larger (and therefore valid) upper bound on the expression. This gives us the final bound on the deviation:

$$\|\Delta\| \le \frac{\gamma}{\lambda(2 + 2\varsigma\|\mathbf{C}_0\|^2)}\epsilon_{\text{SHC}} = \frac{\gamma}{2\lambda + 2\lambda\varsigma\|\mathbf{C}_0\|^2}\epsilon_{\text{SHC}}$$

This final expression shows that the deviation between the ASHC and SHC weights is controlled by the SHC residual error, scaled by a factor that depends on the regularization parameters λ and ς. □

Corollary 1.1. *The primary driver of the deviation $\|\Delta\|$ between the ASHC and SHC weights is the SHC residual error ϵ_{SHC}. The terms λ and $\lambda_{\min}(M)$ appear in the denominator and therefore act only as scaling factors: stronger regularization (λ large) or greater curvature of the Hessian ($\lambda_{\min}(M)$ large) shrink the deviation, but cannot generate deviation on their own. In contrast, ϵ_{SHC} enters multiplicatively in the numerator. Thus, if the SHC fit is already tight (so that ϵ_{SHC} is small), the deviation $\|\Delta\|$ is necessarily small. Conversely, if SHC exhibits poor fit, then ϵ_{SHC} is large, and ASHC permits correspondingly larger deviations, though still bounded by the regularization terms. In this way, ASHC improves stability without moving far from the baseline SHC solution unless the SHC fit leaves room for meaningful improvement.*

To analyze the ASHC estimator relative to the SHC estimator, we aim to bound how much the ASHC predictions can deviate given a change in the weight vector. We do this using the operator norm of the donor matrix $\widetilde{\mathbf{Y}}$, which measures the largest possible amplification of weight deviations into prediction deviations.

Practically, this bound is useful because it quantifies the worst-case sensitivity of the ASHC predictions to perturbations from the SHC weights, providing a guarantee on prediction stability.

Lemma 1.2. *Under Assumption (A4), the error from deviating off SHC weights satisfies*

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|,$$

where $\Delta = \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$*.*

Proof. We begin by recalling the definition of the operator (spectral) norm of a matrix. For a real matrix $A \in \mathbb{R}^{m\times N}$,

$$\|A\| \;:=\; \sup_{\mathbf{v}\in\mathbb{R}^N\setminus\{0\}} \frac{\|A\mathbf{v}\|}{\|\mathbf{v}\|}.$$

This means $\|A\|$ measures the largest possible multiplicative factor by which A can increase the Euclidean norm of a vector.

Now let $A = \widetilde{\mathbf{Y}}$ and $\mathbf{v} = \Delta$, where $\Delta = \mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}} \in \mathbb{R}^N$ is the deviation of the ASHC weight vector from the SHC weight vector.

By the definition of the operator norm, we have

$$\frac{\|\widetilde{\mathbf{Y}}\Delta\|}{\|\Delta\|} \;\leq\; \|\widetilde{\mathbf{Y}}\|, \quad \text{for all } \Delta \neq \mathbf{0}.$$

Multiplying through by $\|\Delta\|$ yields

$$\|\widetilde{\mathbf{Y}}\Delta\| \;\leq\; \|\widetilde{\mathbf{Y}}\|\,\|\Delta\|.$$

Finally, Assumption (A4) gives us the explicit bound

$$\|\widetilde{\mathbf{Y}}\| \;\leq\; \gamma,$$

so that

$$\|\widetilde{\mathbf{Y}}\Delta\| \;\leq\; \gamma\|\Delta\|.$$

In words: although ASHC may choose weights that differ from SHC, the resulting perturbation in the reconstructed donor outcomes is controlled in magnitude, with the constant of proportionality γ coming directly from the operator norm bound in Assumption (A4). ☐

Remark 1.1. *This lemma directly links the deviation in the weights,* $\|\Delta\|$*, to the change in the fitted donor outcomes,* $\|\widetilde{\mathbf{Y}}\Delta\|$*. It uses the operator norm of the donor matrix* $\widetilde{\mathbf{Y}}$*. The operator norm,* γ*, measures the maximum possible "stretching" effect that the donor matrix can have on any vector. It tells us that even if the ASHC weights differ from the SHC weights, the resulting change in the fitted donor outcomes is bounded. Specifically, it cannot exceed the magnitude of the weight deviation multiplied by* γ*. This is important because it ensures that the model is well-behaved. If the donor matrix had a very large operator norm, a small change in the weights could lead to a huge, uncontrolled change in the fitted outcomes. By bounding this norm with* γ*, the lemma guarantees that the impact of weight deviations on the fitted outcomes is predictable and proportional.*

Lemma 1.3. *Under (A3), for horizon* $k > 0$*,*

$$|\ell(T_0+k) - \ell^{SHC}(T_0+k)| \;\leq\; b_\epsilon(H,k) \;:=\; \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

Proof. See Proposition 1 of Chen et al. [2024]. $\square$

Remark 1.2. *Lemma 1.3 formalizes the idea that the SHC predictor inherits the smoothness of the latent trend $\ell(\cdot)$. Because SHC is built from overlapping donor segments that reflect the same underlying smooth dynamics, the approximation reproduces the first H terms of the Taylor expansion around T_0. The only discrepancy comes from the $(H+1)$-st order remainder, which is small when the latent trend is sufficiently smooth. Thus, the difference between ℓ and ℓ^{SHC} grows only polynomially in the forecast horizon k, and the error is of order $|k|^{H+1}$ rather than linear or quadratic in k. This ensures that SHC extrapolates the smooth component accurately over short to moderate horizons.*

2 Finite Sample Bound

Proposition 2.1. *Under assumptions (A1)–(A5), the ASHC prediction error is bounded by*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \;\le\; \epsilon_{SHC}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta.$$

Proof. We want to control the error of the ASHC estimator, given by

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\|.$$

Directly analyzing this is difficult because the weights $\mathbf{w}^{\mathrm{ASHC}}$ are defined through a penalized optimization problem. To make progress, we first recall the triangle inequality, which states that for any vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\|\mathbf{a} + \mathbf{b}\| \le \|\mathbf{a}\| + \|\mathbf{b}\|.$$

This allows us to introduce the SHC weights into the expression and split the ASHC error into two pieces: the SHC prediction error, which we already know how to bound, and the extra error due to the deviation of ASHC from SHC.

Let $\Delta := \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}$ denote the difference between the two weight vectors. Then we can write

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| = \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + \widetilde{\mathbf{Y}}(\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}})\|.$$

Applying the triangle inequality gives

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\le\; \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\| + \|\widetilde{\mathbf{Y}}(\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}})\|.$$

Since $\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}} = -\Delta$, the second term becomes $\|\widetilde{\mathbf{Y}}\Delta\|$. Denoting the SHC error by

$$\epsilon_{\mathrm{SHC}} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\|,$$

we obtain

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\le\; \epsilon_{\mathrm{SHC}} + \|\widetilde{\mathbf{Y}}\Delta\|.$$

The challenge now reduces to bounding $\|\widetilde{\mathbf{Y}}\Delta\|$. For this, we use the operator norm. Recall that for a matrix A,

$$\|A\| := \sup_{\mathbf{v}\neq 0} \frac{\|A\mathbf{v}\|}{\|\mathbf{v}\|}.$$

This definition guarantees that for any vector $\mathbf{v}$,

$$\|A\mathbf{v}\| \le \|A\| \cdot \|\mathbf{v}\|.$$

Taking $A = \widetilde{\mathbf{Y}}$ and $\mathbf{v} = \Delta$, we have

$$\|\widetilde{\mathbf{Y}}\Delta\| \le \|\widetilde{\mathbf{Y}}\| \cdot \|\Delta\|.$$

By Assumption (A4), the operator norm of $\widetilde{\mathbf{Y}}$ is bounded by γ, so

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|.$$

To proceed, we need a bound on the size of the deviation Δ. Lemma 1.1 provides exactly this, showing that

$$\|\Delta\| \leq \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Substituting this into the previous inequality, we obtain

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Finally, we incorporate the other two sources of error. Lemma 1.3 bounds the bias due to imperfect smoothness of the latent trend by

$$b_\epsilon(H,k) = \frac{2\epsilon|k|^{H+1}}{(H+1)!},$$

and Assumption (A2) ensures that the noise term contributes at most δ. Adding these yields

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} + b_\epsilon(H,k) + \delta.$$

It remains only to factor out ϵ_{SHC} from the first two terms:

$$\epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} = \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right).$$

Thus we conclude that

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta,$$

which completes the proof.

□

Remark 2.3. *The proof explicitly shows how each component contributes to the final bound: (1) the SHC baseline error, (2) the ASHC deviation penalty controlled by λ and ς, (3) the smoothness bias from Taylor approximation, and (4) the bounded noise. Factoring ϵ_{SHC} at the end highlights how ASHC scales the SHC error by at most a small multiplicative factor depending on the regularization strength. It is worth emphasizing that the ASHC estimator is guaranteed to perform at least as well as SHC within the evaluation window, since its objective explicitly penalizes deviations from the SHC solution. The bound above quantifies how much its predictions may differ, depending on the donor matrix and the penalty strength. However, this guarantee applies in-sample. Out-of-sample performance in the post-treatment period depends on smoothness of the latent trend and the boundedness of noise, as reflected in the $b_{\epsilon(H,k)}$ and δ terms.*

Lemma 2.4 (Post-treatment smoothness bias and noise)**.** *Under Assumptions (A2) and (A3), the approximation error in the post-treatment period satisfies*

$$\|\mathbf{y}_{post} - \mathbf{Y}_{post}\mathbf{w}^*\| \leq b_{\epsilon,post}(H,k) + \delta_{post},$$

where $b_{\epsilon,post}(H,k)$ is a smoothness bias term similar to the pre-treatment case, and δ_{post} bounds the noise in the post-treatment outcomes.

Proof. Recall from Assumption (A1) that the treated unit's latent potential outcome $\ell(t)$ is $H+1$ times differentiable, and its $(H+1)$-th derivative is bounded by ϵ. By a Taylor expansion argument, the latent outcome over the post-treatment period can be approximated by the linear combination of donor segments with a bias bounded by the smoothness term $b_{\epsilon,\text{post}}(H,k)$.

Additionally, from Assumption (A2), the noise vector $\boldsymbol{\eta}_{\text{post}}$ satisfies $\|\boldsymbol{\eta}_{\text{post}}\| \leq \delta_{\text{post}}$.

Thus,

$$\mathbf{y}_{\text{post}} = \mathbf{Y}_{\text{post}}\mathbf{w}^* + \boldsymbol{\eta}_{\text{post}} + \text{smoothness bias},$$

where the smoothness bias norm is at most $b_{\epsilon,\text{post}}(H,k)$.

Applying the triangle inequality,

$$\|\mathbf{y}_{\text{post}} - \mathbf{Y}_{\text{post}}\mathbf{w}^*\| \leq \|\boldsymbol{\eta}_{\text{post}}\| + b_{\epsilon,\text{post}}(H,k) \leq \delta_{\text{post}} + b_{\epsilon,\text{post}}(H,k).$$

This completes the proof. □

Assumption 2.6 (Post-treatment operator norm)**.** *There exists $\gamma_{post} > 0$ such that*

$$\|\mathbf{Y}_{post}\| \leq \gamma_{post}.$$

Lemma 2.5 (Weight deviation between oracle and SHC)**.** *Under Assumptions (A1)–(A5), the difference between the oracle weights $\mathbf{w}^*$ and the SHC weights $\mathbf{w}^{SHC}$ is bounded by*

$$\|\mathbf{w}^* - \mathbf{w}^{SHC}\| \leq \eta,$$

for some $\eta > 0$ depending on the data and model.

Proof. This follows from the stability and consistency properties of the SHC estimator under Assumptions (A1)–(A5). Since $\mathbf{w}^{\text{SHC}}$ solves a constrained least squares problem minimizing

$$\left\|\widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{eval}}\mathbf{w}\right\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2,$$

and $\mathbf{w}^*$ satisfies the noiseless model $\widetilde{\mathbf{y}}_{\text{eval}} = \widetilde{\mathbf{Y}}_{\text{eval}}\mathbf{w}^* + \boldsymbol{\eta}$ with $\|\boldsymbol{\eta}\| \leq \delta$, standard results on constrained least squares imply $\mathbf{w}^{\text{SHC}}$ is close to $\mathbf{w}^*$ within a ball whose radius depends on δ and model parameters.

More explicitly, since the problem is strictly convex and well-conditioned by assumption (due to the regularization and eigenvalue structure),

$$\|\mathbf{w}^* - \mathbf{w}^{\text{SHC}}\| \leq \eta,$$

where η depends on the noise bound δ, the regularization parameter ς, and the spectral properties of $\widetilde{\mathbf{Y}}_{\text{eval}}$.

A detailed exact expression can be derived by comparing the KKT conditions for $\mathbf{w}^*$ and $\mathbf{w}^{\text{SHC}}$, but for the purposes here, it suffices to assert the existence of such η.

□

Lemma 2.6 (Total weight deviation bound)**.** *We have*

$$\|\mathbf{w}^* - \mathbf{w}^{ASHC}\| \leq \|\mathbf{w}^* - \mathbf{w}^{SHC}\| + \|\mathbf{w}^{SHC} - \mathbf{w}^{ASHC}\|.$$

Proof. This is a direct application of the triangle inequality for vector norms. For any vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in a normed vector space,

$$\|\mathbf{a} - \mathbf{c}\| \leq \|\mathbf{a} - \mathbf{b}\| + \|\mathbf{b} - \mathbf{c}\|.$$

Setting $\mathbf{a} = \mathbf{w}^*$, $\mathbf{b} = \mathbf{w}^{\text{SHC}}$, and $\mathbf{c} = \mathbf{w}^{\text{ASHC}}$ yields the stated result. □

3 Asymptotic Results

Theorem 3.1. *Under (A1)–(A5), as $m \to \infty$ and $H \to \infty$, for fixed $k > 0$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| \longrightarrow \delta.$$

If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| \xrightarrow{p} 0.$$

Proof. Starting from Proposition 2.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \le \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Behavior of $\epsilon_{\mathbf{SHC}}$. By (A3), $\ell(t)$ is $(H+1)$-times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\mathrm{SHC}} \to 0 \quad \text{as } m \to \infty,\ H \to \infty.$$

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Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\mathrm{SHC}} \to 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}}\, \epsilon_{\mathrm{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H,k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \le \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Thus

$$b_\epsilon(H,k) \le \frac{2\epsilon}{(H+1)!} |k|^{H+1} \to 0 \quad \text{as } H \to \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \le \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2, then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,

$$\limsup_{m \to \infty} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \le \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. □

Corollary 3.2. *Under (A1)–(A5), the ASHC estimator is asymptotically unbiased up to δ. If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy σ^2 and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \to \infty$,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{ASHC}\| = O_p\left(m^{-H/(2H+3)}\right).$$

For large H, this approaches the parametric rate $O_p(m^{-1/2})$.

Proof. **Part 1: Consistency.** From Theorem 3.1, the error converges to δ, or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Proposition 2.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \le \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H,k) + \delta.$$

Step 1. Bias term. From Lemma 1.3,

$$b_\epsilon(H,k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is

$$O_p\big((m h_m)^{-1/2}\big).$$

Step 3. SHC fit error. Combining,

$$\epsilon_{\mathrm{SHC}} = O_p\big(h_m^H + (m h_m)^{-1/2}\big).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (m h_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,

$$\epsilon_{\mathrm{SHC}} = O_p\big(m^{-H/(2H+3)}\big).$$

Thus the total error is

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| = O_p\big(m^{-H/(2H+3)}\big).$$

Step 6. Near-parametric rate. As $H \to \infty$,

$$\frac{H}{2H+3} \to \tfrac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. □

Remark 3.4. *The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \le \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((m h_m)^{-1/2})$. This explains the appearance of the nonparametric rate.*

References

Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: https://ssrn.com/abstract=4995085, September 2024. URL http://dx.doi.org/10.2139/ssrn.4995085.